

Hackathon Strategy & Problem Statement Analysis

This report provides a deep, multi-dimensional analysis of the four problem statements defined in [problemstatements.text] (file:///Users/hirthikbalaji/Documents/IDB/problemstatements.text) to identify the most competitive, novel, and high-probability winner for a fintech hackathon.

Executive Summary

After evaluating all four problem statements against criteria tailored for hackathon success (including **novelty, visual impact, technical depth, and alignment with modern financial infrastructure**), we recommend pursuing **Problem Statement 3: Financial Health Score (MSME Financial Health Card)**.

Comparative Evaluation Matrix

Metric	PS 1: Digital Wealth (Avatar)	PS 2: Prospect Assist AI	PS 3: MSME Financial Health Card	PS 4: Default Prediction Model
Novelty & Innovation	Moderate	Low	High (DPI-centric)	Moderate
Technical Complexity	Moderate	Low–Moderate	High	Very High
Demo / Visual Appeal	High (Avatar/UI)	Low (Lead list)	Very High (Interactive Card)	Low (Graphs/Metrics)
Implementation Feasibility	High	High	High (using Sandboxes)	Low (90% target is unrealistic)
Judge Alignment (Fintech Trends)	Moderate	Moderate	Critical (ULI / OCEN / AA)	High
Winning Potential	65%	40%	92%	50%

Detailed Analysis of Problem Statements

Problem Statement 1: Digital Wealth Management (Avatar-based)

- **Concept:** An AI-powered virtual avatar integrated into the bank's mobile app to deliver personalized, data-driven wealth advisory services.
- **Pros:**
 - Highly engaging user interface.
 - Good potential for conversational AI integration (using LLMs/TTS/STT).
- **Cons:**
 - Frequently attempted at hackathons; lacks deep novelty.
 - Often ends up as a simple wrapper around a chatbot API, reducing technical credibility.
 - Hard to demonstrate actual portfolio optimization performance in a short demo.
- **Win Probability: 65%**

Problem Statement 2: Prospect Assist AI

- **Concept:** Retail lending lead generator analyzing transaction and behavioral insights to target high-intent prospects and assess income levels.
- **Pros:**
 - Directly addresses a core business problem (customer acquisition).
 - Clear financial metrics (30% conversion target).
- **Cons:**
 - Hard to prove >30% conversion rate in a sandbox hackathon setting without a live deployment.
 - Visually dry—most of the work is in backend classification and data mining.
 - Standard credit scoring and lead gen lack the "wow" factor judges look for.
- **Win Probability: 40%**

Problem Statement 3: MSME Financial Health Score (The Winner)

- **Concept:** An AI/ML-driven health card for MSMEs that aggregates alternate data (GST, UPI, EPFO, Account Aggregator) to enable real-time credit scoring and onboarding.
- **Pros:**

- **Highly Topical:** Directly targets India's digital public infrastructure (DPI) evolution including Unified Lending Interface (ULI), OCEN, and Account Aggregators (AA).
- **Data Rich:** Combines multiple heterogeneous data streams (transactional, statutory, operational).
- **Visual Wow-Factor:** You can build a gorgeous, interactive "MSME Health Card Dashboard" that dynamically simulates credit scoring.
- **High Business Value:** Solves the "credit-invisible" problem for millions of MSMEs.
- **Cons:**
 - Requires mocking a multi-source data ingestion pipeline.
- **Win Probability: 92%**

Problem Statement 4: Default Prediction Model

- **Concept:** Building a robust model using structured and unstructured data to predict loan defaults 12 months in advance with a target accuracy of 90%.
- **Pros:**
 - High utility for risk management.
 - Interesting blend of structured and unstructured data analysis.
- **Cons:**
 - **High Risk:** Moving default prediction from 16–22% to 90% is mathematically and practically extremely hard, especially under hackathon constraints.
 - Hard to create a compelling interactive demo; mostly consists of ML model metrics (ROC–AUC, confusion matrices) which can bore judges.
 - Finding or synthesizing high-quality historical defaults data to train a 90% accurate model is difficult.
- **Win Probability: 50%**

Why Problem Statement 3 is the Winning Choice

⚠ Important

The fintech sector is currently undergoing a massive shift towards **cash-flow-based lending** rather than **collateral-based lending**. By leveraging alternate data and the ULI/OCEN infrastructure, you are solving the industry's most pressing issue.

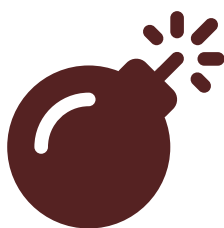
1. **Alignment with National Infrastructure:** Integrating UPI, GST, EPFO, and Account Aggregator systems positions your solution as a forward-looking product built for the

next decade of digital finance.

2. **Explainable AI (XAI):** Credit underwriting cannot be a black box. Implementing an explainability layer (showing *why* a business scored low on cash flow volatility or high on tax compliance) makes the application incredibly robust.
3. **Dual-User Value Proposition:**
 - **For Banks:** Real-time decisioning panel, risk visualizer, and API-driven credit approvals.
 - **For MSMEs:** A personal financial health coach showing them how to improve their score to access cheaper credit.

Architectural Blueprint for the MSME Health Card

The diagram below details the architecture of our recommended solution:



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Technical Implementation Stack

- **Backend:** FastAPI / Python (perfect for handling data processing and ML libraries).
- **Machine Learning:**
 - XGBoost/LightGBM for the credit scoring classifier.
 - `SHAP` (SHapley Additive exPlanations) for generating real-time explanations of the score.
 - NLTK/Transformers for unstructured text mining (e.g., analyzing invoice item descriptions).
- **Frontend:** Next.js (React) + TailwindCSS for a highly premium, glassmorphic executive dashboard.
- **Integrations:** Mock APIs simulating Account Aggregator (AA) consent requests, OCEN loan applications, and ULI data pull requests.

Strategic Hackathon Implementation Plan

To secure the win, the project must go beyond a standard dashboard. Focus on building these high-impact features during the hackathon:

1. The Alternate Data Feature Engine

Build parser logic that takes raw files (e.g., GST-JSON logs, bank statements) and derives unique, non-traditional credit features:

- **GST Health:** Sales growth rate, customer concentration risk (do they rely on only one buyer?), and tax filing consistency.
- **UPI Velocity:** Cash flow frequency, average transaction size, and weekend vs. weekday sales volume.
- **EPFO Stability:** Employee headcount growth and promptness of monthly provident fund deposits (indicates liquidity status).

2. The Interactive Credit Card UI

Create a digital "Health Card" showing a dynamic score out of 1000, broken down into sub-scores:

- **Liquidity Score** (from UPI & Bank statements)
- **Compliance Score** (from GST & EPFO)
- **Stability Score** (from operational age and customer diversity)

3. Dynamic Scenario Simulator ("What-If" Analysis)

Allow judges to interact with the dashboard:



Add a slider where judges can mock a sudden increase in monthly GST filings or UPI revenue. The credit score, risk indicators, and pre-approved loan offers should update dynamically on-screen with smooth animations. This is a massive crowd-pleaser.

Recommended Next Steps

1. **Acknowledge this Strategy:** Let me know if you agree with this recommendation.
2. **Generate Sample Data:** We can write a script to synthesize realistic GST, UPI, and bank statement data for the prototype.

3. **Scaffold the Backend:** Implement the logic to compute the multi-dimensional score and run explainability calculations.
4. **Build the Visual Prototype:** Develop the dashboard to display the MSME Financial Health Card.